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KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY
KUMASI

COLLEGE OF ENGINEERING

B.Sc. (Engineering) Second Semester Examination, 2014
Second Year

ME 296 MECHANICAL ENGINEERING LABORATORY II

Answer ALL questions **TIME ALLOWED 2 Hrs.**

Q1.

A. From your characteristic study on Bourdon tube pressure gauge explain the principle of how a Bourdon tube operates.

B. Sketch a Bourdon tube pressure gauge showing the mechanical details and label the following:
   i. Receiver block
   ii. Chassis plate
   iii. Secondary chassis plate
   iv. Stationary end of bourdon tube
   v. Movable end of bourdon tube

C. The indicator spring setup includes a small piston, drum with a piece of card or paper, Stylus, a cord and calibrated spring.

   i. Explain the function of the small piston with the aid of a sketch
   ii. State what happens to the card on the drum when you pull the cord
   iii. State what happens to the card on the drum when the cord is connected to a reciprocating mechanism

Q2.

i) Sketch and fully label the Screw Jack Apparatus setup

ii) List all the apparatus your group used for the experiment on the screw jack experiment

iii) List at least five (5) precautions you and group observed when your group conducted experiment on the Screw Jack.

iv) State the laboratory you conducted the torsional pendulum experiment

Q3.

i) State fully the objective of conducting the experiment on "Relationship between Temperature and Pressure of Saturated Steam"

ii) State THREE precautions your group observed in conducting the experiment stated in Q3(i)

iii) List the apparatus your group used to conduct the experiment on "Relationship between Temperature and Pressure of Saturated Steam".

Q4.

A. Your group conducted an experiment to determine the property of oil. Sketch and label the experimental setup.

B. You have conducted characteristic study on "Bomb Calorimeter".

i. state the main purpose of the experiment in your laboratory which involves the application of bomb calorimeter

ii. Sketch the experimental graph associated with Q4 (Bi).

iii. From Q4(Bii) explain Beckmann Reading

iv. Name the experimental setup in figure Q4B and label it fully

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ME 262 THEORY OF MACHINES I

APRIL, 2014 THREE HOURS

Attempt all questions.

1. The shaft shown in Figure 1 is to be balanced by placing correction masses in the planes L and R. The weights of the three masses m_1, m_2 *and* m_3 *are respectively*, are respectively 4, 3 and 4 kg. The dimensions in cm are $R_1 = 5, R_2 = 4, R_3 = 5$, $a = 1$, $b = e = 8, c = 10$ *and* $d = 12$. Calculate the magnitude of the corrections and their angular locations.

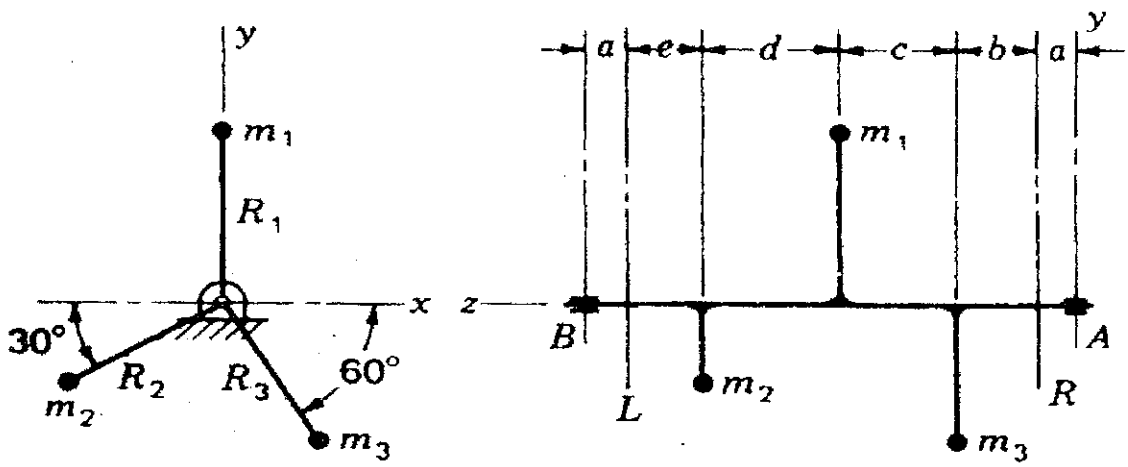


Figure 1

2. i. Explain briefly the following terms;

   a. kinematic inversion

b. mobility

c. coupler curve

ii. List six industrial applications of the four-bar linkage.

iii. List the six lower kinematic pairs. Revolute, Prismatic, Screw, Cylindrical, Spherical, Planar.

iv. Give at least three types of mechanisms with sketches for transmitting motions, showing two applications of each type so given.

3. The simple slider-crank mechanism shown in Figure 2 consists of a crank OA, slider block B, and connecting link AB. If the linear velocity and acceleration of the block B to the left are 10 m/s and 800 m/s^2 respectively. OA=0.1 m and AB=0.4 m. Determine the velocity and acceleration of point A at this instant.

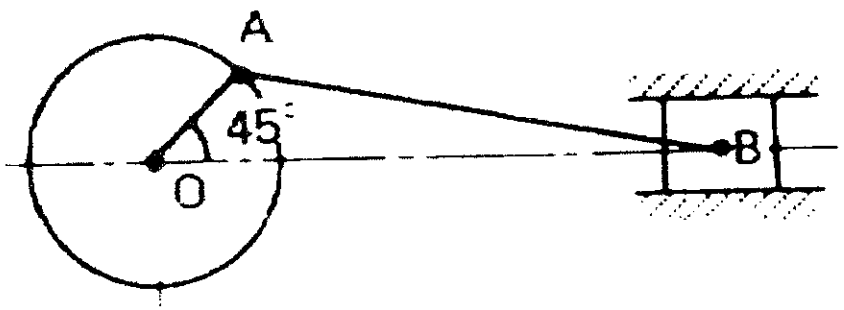


Figure 2

Hint: $[v_B = v_A + (v_{B/A})_{xyz} + \Omega \times r_{B/A}\ ;\ a_B = a_A + (a_{B/A})_{xyz} + \dot{\Omega} \times r_{B/A} + 2\Omega \times (v_{B/A})_{xyz} - \Omega^2 r_{B/A}]$

4. The Figure below shows a reverted planetary train. Gear 2 is fastened to its shaft and is driven at 450 rpm in anti-clockwise direction. Gears 4 and 5 are planet gears which are joined but are free to turn on the shaft carried by the arm. Gear 6 is stationary. Find the speed and direction of rotation of the arm.

$$\frac{\omega_{6/3}}{\omega_{2/3}} = -\frac{N_2}{N_4} \cdot \frac{N_5}{N_6} = \frac{\omega_6 - \omega_3}{\omega_2 - \omega_3}$$

$$-\frac{20}{30} \cdot \frac{16}{34} = \frac{0 - \omega_3}{450 - \omega_3}$$

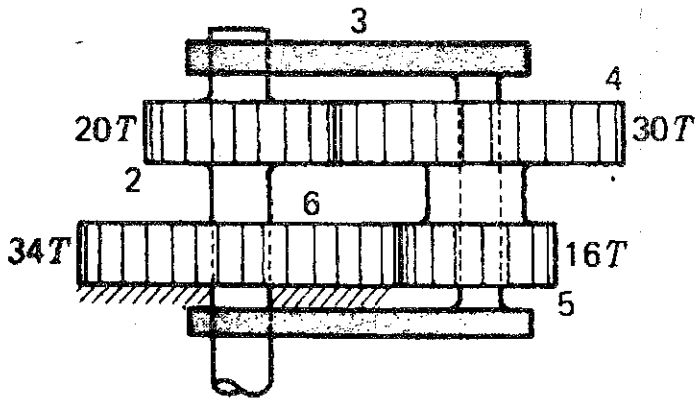


Figure 3

5. In Figure 4 a block of mass m = 50 kg is supported by dampers of damping coefficients c_1 = 20 N s/m and c_2 = 50 N s/m, a spring of stiffness k= 120 N/m on a smooth horizontal surface. If the block is displaced x = 50 mm to the right of the equilibrium position and released from rest.

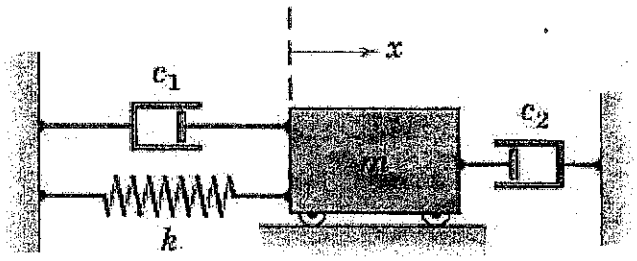


Figure 4

(i) Obtain the free-body diagram and the equation of motion

(ii) Determine the;

(a) natural frequency

(b) damping ratio

(c) damped natural frequency

(iii) Solve and sketch the response

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